



Online Data Storage

JSON, MySQL & Rest PHP API

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2025-2026

Application Programming Interface (API)

An application programming interface (API) is an interface that defines interactions between multiple software applications or mixed hardware-software intermediaries. It defines the kinds of calls or requests that can be made, how to make them, the data formats that should be used, the conventions to follow, etc. In the mobile app development lifecycle, communicating with the server (via API) to fetch or store data is one of the basic needs.

If you were to interact with an API, what format do you think the response would be in? It would probably — or almost definitely — be in JSON format. First of all, let's understand JSON;

JSON Format

JSON is the standard for transferring typed data from one point to another point. If you query an API, or even want to save settings in your app, you're going to have to use JSON sooner or later. So, whenever you work with API, we usually get responses in JSON format. JSON format looks somewhat like the below:

```
{
  "name": "Abhishek",
  "age": 22,
  "technology": [
    {
      "name": "Flutter",
      "experience": 2
    },
    {
      "name": "GIT",
      "experience": 2
    }
  ]
}
```

Parsing JSON

Nowadays JSON is one of the most popular data-interchange formats. For example, very likely you'll have to deal with HTTP requests returning a json-encoded string. Flutter has an incredible support for this format as it allows you to convert objects from and to JSON very quickly.

```
{
  "id": 1,
  "name": "Alberto",
}
```

What's inside curly brackets is called **json object** and it's always made up of a string, which is the key of the field, and a value which can be a string, a number, a boolean, a list, null or another object. This is another example with a JSON array (or list):

```
[
  {
    "id": 1,
    "name": "Alberto",
  },
  {
    "id": 2,
    "name": "Patrick",
  }
]
```

Lists are simply collections of objects. In Flutter you can decide between manually parsing strings, using facilities from `import "dart:convert"`, or using automatic code generation, which does most of the work automatically. Both ways are fine but you'll see that code generation simplifies maintenance a lot, especially for large JSON objects.

Example: Local json file from asset folder in flutter

In this Flutter example, We will learn **how to read json file in flutter** & [parse local JSON file](#) in flutter & the **Display the JSON file data** in flutter listview builder.

1. Create a new flutter project

Firstly, need to create a new flutter project or simply only any existing flutter project where you want to implement read local JSON file in flutter.

In my case, I make use of android studio to develop flutter application.

creating new flutter project: `File > New > New Flutter Project.`

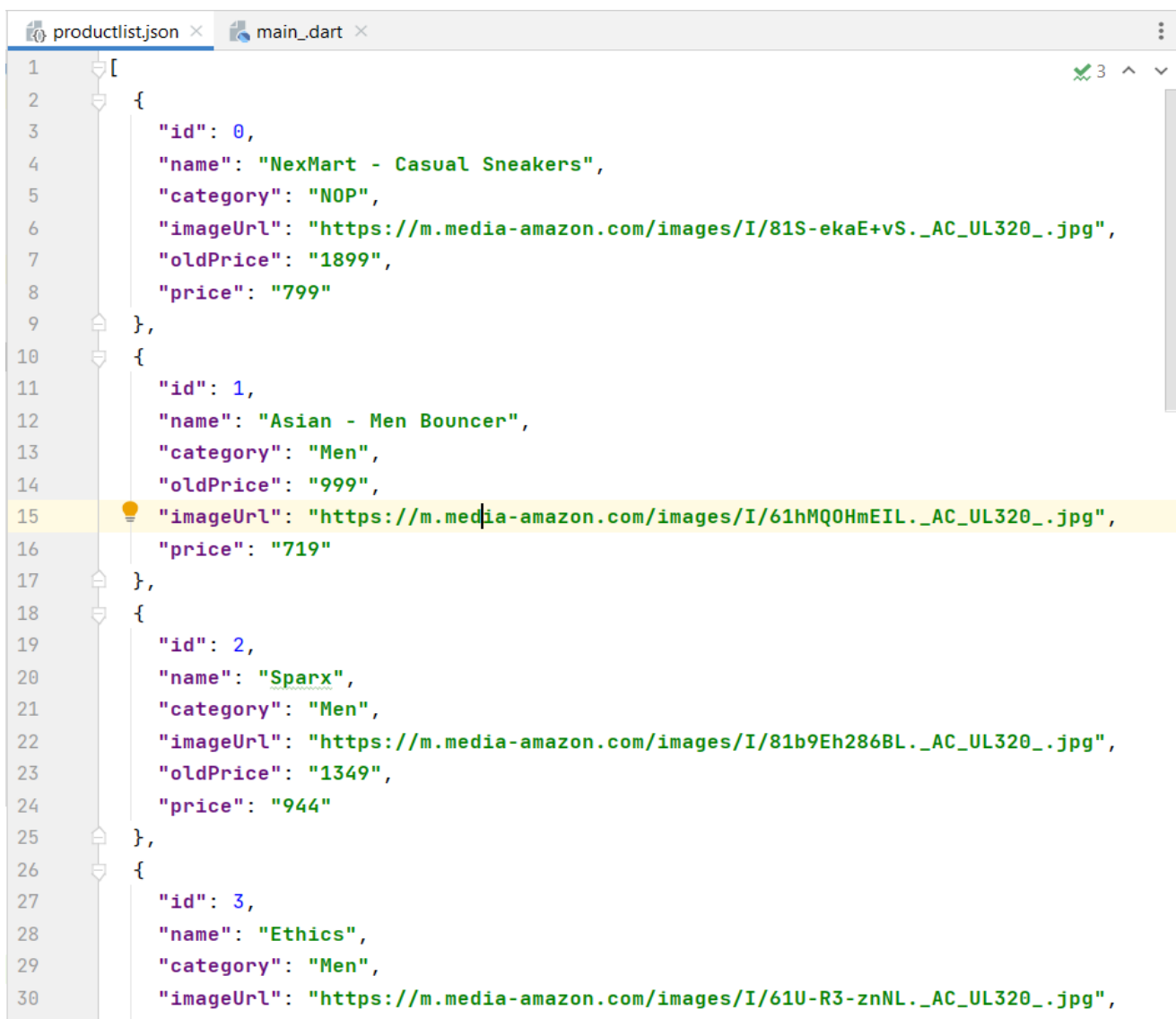
Give a good name & package name to project and finish. Done your flutter project is ready.

2. How to add JSON file in flutter project as asset file

I have created a JSON file by name **ProductItem.json** which has a list of 10 shoes products (id, name, category, imageUrl, oldPrice & price of each item).you can download the json file from below link or simply create you own json file.

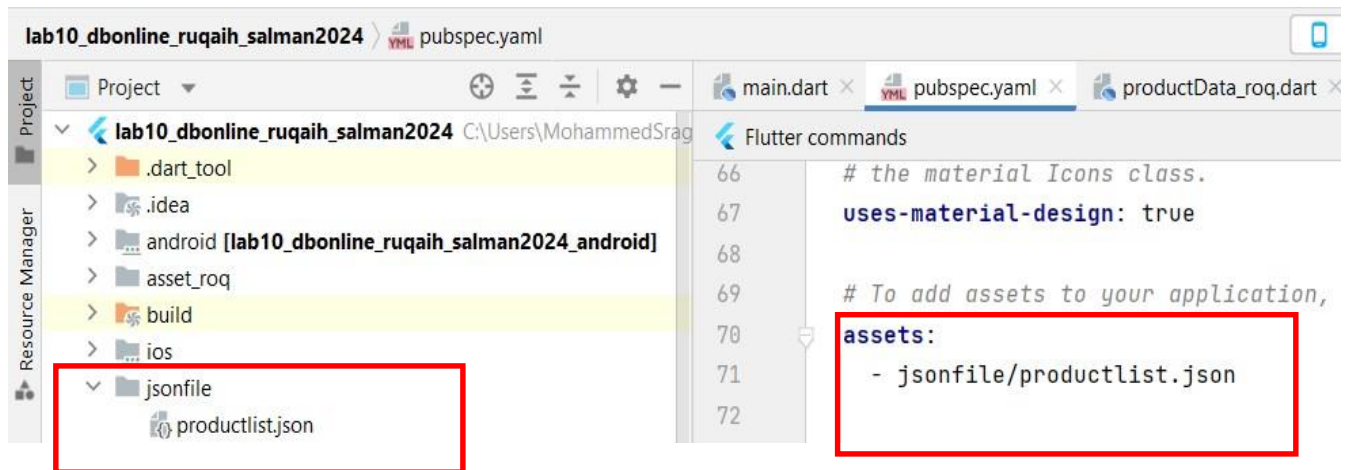
<https://protocoderspoint.com/jsondata/productlist.json>

ProductItem.json file



```
1  [
2    {
3      "id": 0,
4      "name": "NexMart - Casual Sneakers",
5      "category": "NOP",
6      "imageUrl": "https://m.media-amazon.com/images/I/81S-ekaE+vS._AC_UL320_.jpg",
7      "oldPrice": "1899",
8      "price": "799"
9    },
10   {
11     "id": 1,
12     "name": "Asian - Men Bouncer",
13     "category": "Men",
14     "oldPrice": "999",
15     "imageUrl": "https://m.media-amazon.com/images/I/61hMQ0HmEIL._AC_UL320_.jpg",
16     "price": "719"
17   },
18   {
19     "id": 2,
20     "name": "Sparx",
21     "category": "Men",
22     "imageUrl": "https://m.media-amazon.com/images/I/81b9Eh286BL._AC_UL320_.jpg",
23     "oldPrice": "1349",
24     "price": "944"
25   },
26   {
27     "id": 3,
28     "name": "Ethics",
29     "category": "Men",
30     "imageUrl": "https://m.media-amazon.com/images/I/61U-R3-znNL._AC_UL320_.jpg",
```

Create a directory under flutter project **jsonfile** & paste the **ProductItem.json** file in **jsonfile** directory. Then add the **ProductItem.json** file path in asset in pubspec.yaml as blow image



3. Create a data model class file

ProductDataModel.dart is used to handle our data that is been used to read from JSON file.

This class has dataTypes & constructor & a method.

ProductDataModel.fromJson is used to assign the data to variable in [MAP json](#) format.

➤ ProductDataModel.dart

```
class ProductDataModel{
//data Type
  int? id;
  String? name;
  String? category;
  String? imageURL;
  String? oldPrice;
  String? price;
// constructor
  ProductDataModel (
    {
      this.id,
      this.name,
      this.category,
      this.imageURL,
      this.oldPrice,
      this.price
    }
  );
//method that assign values to respective datatype vairables
  ProductDataModel.fromJson (Map<String,dynamic> json)
  {
    id = json['id'];
```

```

    name = json['name'];
    category = json['category'];
    imageURL = json['imageUrl'];
    oldPrice = json['oldPrice'];
    price = json['price'];
  }
}

```

4. Reading JSON file & display the data in listview

Explanation of what is done in **main.dart** file.

Step1: Import services.dart as rootBundle

```
import 'package:flutter/services.dart' as rootBundle;
```

This will help us in reading json file.

Step2: Create Method to read json file in flutter

```

Future<List<ProductDataModel>> ReadJsonData() async {
  //read json file
  final jsondata = await
rootBundle.rootBundle.loadString('jsonfile/productlist.json');
  //decode json data as list
  final list = json.decode(jsondata) as List<dynamic>;
  //map json and initialize using DataModel
  return list.map((e) => ProductDataModel.fromJson(e)).toList();
}

```

Step3: FutureBuilder to parse & display data in listview

```

FutureBuilder(
  future: ReadJsonData(),
  builder: (context, data) {
    if (data.hasError) {
      return Center(child: Text("${data.error}"));
    } else if (data.hasData) {
      var items = data.data as List<ProductDataModel>;
      return ListView.builder(
        itemCount: items == null ? 0 : items.length,
        itemBuilder: (context, index) {
          return Card(
            elevation: 5,
            margin: EdgeInsets.symmetric(horizontal: 10, vertical: 6),
            child: Container(

```

```

padding: EdgeInsets.all(8),
child: Row(
  mainAxisAlignment: MainAxisAlignment.center,
  crossAxisAlignment: CrossAxisAlignment.center,
  children: [
    Container(
      width: 50,
      height: 50,
      child: Image(
        image:
          NetworkImage(items[index].imageUrl.toString()),
        fit: BoxFit.fill,
      ),
    ),
    Expanded(
      child: Container(
        padding: EdgeInsets.only(bottom: 8),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Padding(
              padding: EdgeInsets.only(left: 8, right: 8),
              child: Text(
                items[index].name.toString(),
                style: TextStyle(
                  fontSize: 16,
                  fontWeight: FontWeight.bold),
              ),
            ),
            Padding(
              padding: EdgeInsets.only(left: 8, right: 8),
              child: Text(items[index].price.toString()),
            ),
          ],
        ),
      ),
    ],
  ),
);
} else {
  return Center(
    child: CircularProgressIndicator(),
  );
}
},
)
)

```

Complete Code - Read JSON file in flutter & display data on screen
 using ListView.Builder
main.dart

```

import 'dart:convert';
import 'package:flutter/cupertino.dart';

```

```

import 'package:flutter/material.dart';

import 'package:flutter/services.dart' as rootBundle;

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Demo',
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: ReadJSON_FuturBuilder_Roq(),
    );
  }
}

class ReadJSON_FuturBuilder_Roq extends StatefulWidget {
  @override
  _ReadJSON_FuturBuilder_RoqState createState() => _ReadJSON_FuturBuilder_RoqState();
}

class _ReadJSON_FuturBuilder_RoqState extends State<ReadJSON_FuturBuilder_Roq> {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: FutureBuilder(
        future: ReadJsonData(),
        builder: (context, data) {
          if (data.hasError) {
            return Center(child: Text("${data.error}"));
          } else if (data.hasData) {
            var items = data.data as List<ProductDataModel>;
            return ListView.builder(
              itemCount: items == null ? 0 : items.length,
              itemBuilder: (context, index) {
                return Card(
                  elevation: 5,
                  margin: EdgeInsets.symmetric(horizontal: 10, vertical: 6),
                  child: Container(
                    padding: EdgeInsets.all(8),
                    child: Row(
                      mainAxisAlignment: MainAxisAlignment.center,
                      crossAxisAlignment: CrossAxisAlignment.center,
                      children: [
                        Container(
                          width: 50,
                          height: 50,
                          child: Image(
                            image:
                              NetworkImage(items[index].imageUrl.toString()),
                            fit: BoxFit.fill,

```

```

    ),
  ),
  Expanded(
    child: Container(
      padding: EdgeInsets.only(bottom: 8),
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          Padding(
            padding: EdgeInsets.only(left: 8, right: 8),
            child: Text(
              items[index].name.toString(),
              style: TextStyle(
                fontSize: 16,
                fontWeight: FontWeight.bold),
            ),
          ),
          Padding(
            padding: EdgeInsets.only(left: 8, right: 8),
            child: Text(items[index].price.toString()),
          ),
        ],
      ),
    ),
  ),
);
} else {
  return Center(
    child: CircularProgressIndicator(),
  );
}
},
);
}

Future<List<ProductDataModel>> ReadJsonData() async {
  final jsondata = await
rootBundle.rootBundle.loadString('jsonfile/productlist.json');
  final list = json.decode(jsondata) as List<dynamic>;
  return list.map((e) => ProductDataModel.fromJson(e)).toList();
}

class ProductDataModel{
  //data Type
  int? id;
  String? name;
  String? category;
  String? imageURL;
  String? oldPrice;
  String? price;

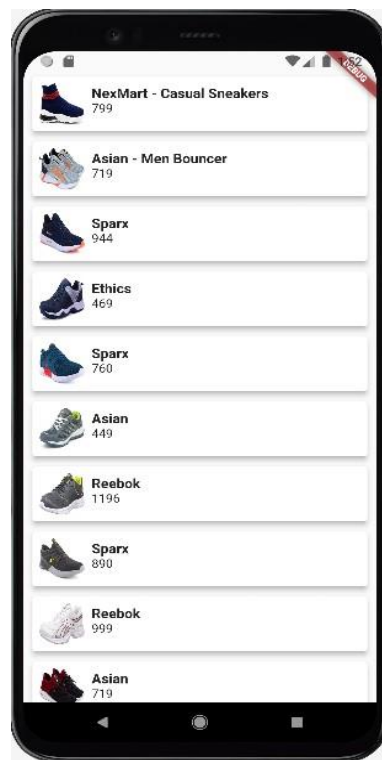
  // constructor

```

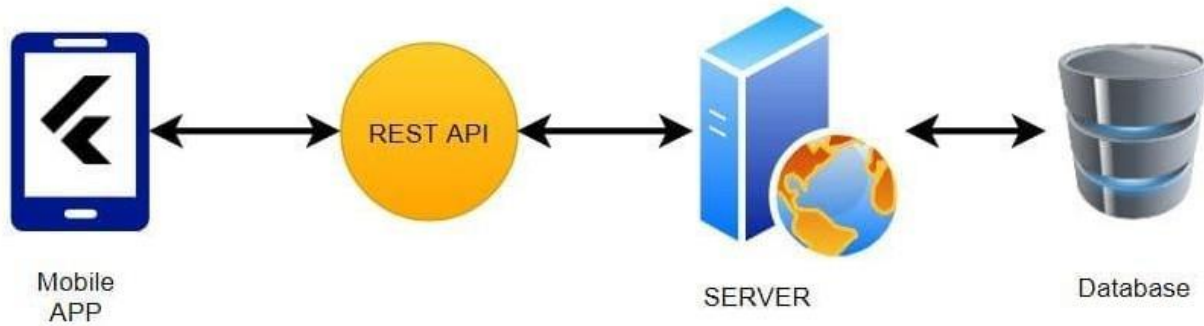
```

ProductDataModel(
{
  this.id,
  this.name,
  this.category,
  this.imageURL,
  this.oldPrice,
  this.price
}
);
//method that assign values to respective datatype vairables
ProductDataModel.fromJson(Map<String,dynamic> json)
{
  id = json['id'];
  name =json['name'];
  category = json['category'];
  imageURL = json['imageUrl'];
  oldPrice = json['oldPrice'];
  price = json['price'];
}
}

```



Rest API in Dart?



Rest APIs are a way to **fetch data from the internet in flutter** or communicate the server from the app and get some essential information from your server to the app. This information can be regarding your app's data, user's data, or any data you want to share globally from your app to all of your users. This HTTP request fetches in a unique [JSON](#) format, and then the information is fetched from the JSON and put in the UI of the app.

Every programming language has a way of some internet connectivity i.e, use this rest API developed on the server and [fetch data from the internet](#).

To use this request feature, Flutter provides **http package** to use http resources. The http package uses **await** and **async** features and provides many high-level methods such as read, get, post, put, head, and delete methods for sending and receiving data from remote locations. These methods simplify the development of REST-based mobile applications.

Work with Rest API in Flutter

There are a few steps that we can follow to easily integrate an API into our Flutter app:

Step 1: Get the API URL and endpoints.

Step 2: Add relevant packages into the app (http, dio, chopper, etc.).

Step 3: Create a constant file that stores URLs and endpoints.

Step 4: Create a model class to parse the JSON.

Step 5: Create a file that handles the API call, and write specific methods to fetch and parse data.

Step 6: Use the data in your app.

Fetching JSON Data using HTTP

Implementation:

We are assuming you already know how to make a project in a Flutter, we will be working on Android Studio.

The following steps are:

Step 1: Create a project in Vs code, And remove the default code.

Step 2: Before writing the code just add the HTTP plugin in your *pubspec yaml* file.

dependencies:

```
http: ^0.13.3
```

OR you can simply add your plugin from the terminal just type this command,

```
flutter pub add http
```

After that run this command for getting the packages,

```
flutter pub get
```

Step 3: In *main.dart* file call the *main()* function , inside it run the *runApp()* method and give it an App (MyApp).

Step 4: Now create a stateful widget with the name 'MyApp'. A stateful widget is a widget that changes their state during runtime, and return the *MaterialApp()*, *MaterialApp* has so many properties , but here we use only 2 or 3, make the *debugBanner: false* , *title : "MyApi"* and in the home property give a widget as you want, we give it *MyApi()*. Now let's create the class *MyApi*, you can work on the same file but here we are working in different files for a different task, so we create a new file *MyApi.dart*.

```
import 'package:flutter/material.dart';  
  
import 'myApiWidget_roq.dart';  
  
void main() {  
  runApp(MyApp());  
}  
  
class MyApp extends StatefulWidget {  
  MyApp({Key? key}) : super(key: key);
```

```

@override
_MyAppState createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: "MyApi",
      debugShowCheckedModeBanner: false,
      home: MyApi(),
    );
  }
}

```

Step 5: In *MyApi.dart* file, make a stateful widget *MyApi* and return the scaffold. In scaffold, there is an *AppBar* and in the *AppBar* we have a *title* : "Fetch Data", and in the body, we have a widget *myApiWidget()*.

```

import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

class MyApi extends StatefulWidget {
  MyApi({Key? key}) : super(key: key);

  @override
  _MyApiState createState() => _MyApiState();
}

class _MyApiState extends State<MyApi> {
  var response;

  Future<List<dynamic>> fetchUsers() async {
    var response =
      await http.get(Uri.parse("https://randomuser.me/api/?results=20"));
    return jsonDecode(response.body)['results'];
  }

  @override
  void initState() {
    response = fetchUsers();
    super.initState();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Fetch Data"),
      ),
      body:
        myApiWidget()
    );
  }
}

```

```
);  
}
```

For data, we have this [JSON](https://randomuser.me/api/?results=20) file in <https://randomuser.me/api/?results=20> link as blow image.

```
{  
  "results": [  
    {  
      "gender": "female",  
      "name": {  
        "title": "Ms",  
        "first": "Vaishnavi",  
        "last": "Bharanya"  
      },  
      "location": {  
        "street": {  
          "number": 2169,  
          "name": "College St"  
        },  
        "city": "Tadepalligudem",  
        "state": "Uttar Pradesh",  
        "country": "India",  
        "postcode": 71453,  
        "coordinates": {  
          "latitude": "6.7354",  
          "longitude": "12.8576"  
        },  
        "timezone": {  
          "offset": "-12:00",  
          "description": "Eniwetok, Kwajalein"  
        }  
      },  
      "email": "vaishnavi.bharanya@example.com",  
      "login": {  
        "uuid": "f2b033ef-a8ce-45bf-85bd-89b97aedfeec",  
        "username": "ticklishleopard658",  
        "password": "punkrock",  
        "salt": "HV6vS0su",  
        "md5": "88dbe1d400434cd2c48c8b16e2b6ce5a",  
        "sha1": "cc3e4481080a1ddc48d0180c0f813f396f138c56",  
        "sha256": "7082799598554b229bf3ee03dcfa339fbel1f0e393acd2bd46e6f92b3e0a31c40"  
      },  
      "dob": {  
        "date": "1984-01-15T03:52:17.606Z",  
        "age": 39  
      },  
      "registered": {  
        "date": "2014-02-26T06:01:56.715Z",  
        "age": 9  
      },  
      "phone": "9989897293",  
      "cell": "9154072251",  
      "id": {  
        "name": "UIDAI",  
        "value": "393870074389"  
      },  
      "picture": {  
        "large": "https://randomuser.me/api/portraits/women/47.jpg",  
        "medium": "https://randomuser.me/api/portraits/med/women/47.jpg",  
        "thumbnail": "https://randomuser.me/api/portraits/thumb/women/47.jpg"  
      },  
      "nat": "IN"  
    },  
    {  
      "gender": "male",  
      "name": {  
        "title": "Mr",  
        "first": "Hugo",  
        "last": "White"  
      },  
      "location": {  
        "street": {  
          "number": 100,  
          "name": "Park Road"  
        },  
        "city": "New York",  
        "state": "New York",  
        "country": "USA",  
        "postcode": 10001,  
        "coordinates": {  
          "latitude": "40.7128",  
          "longitude": "-74.0060"  
        },  
        "timezone": {  
          "offset": "-04:00",  
          "description": "Atlantic Standard Time"  
        }  
      },  
      "email": "hugo.white@example.com",  
      "login": {  
        "uuid": "12345678-90ab-cdef-ghij-klmnopqrstuv",  
        "username": "hugo.white",  
        "password": "password123",  
        "salt": "salt123",  
        "md5": "md5hash",  
        "sha1": "sha1hash",  
        "sha256": "sha256hash"  
      },  
      "dob": {  
        "date": "1980-01-01T00:00:00.000Z",  
        "age": 43  
      },  
      "registered": {  
        "date": "2010-01-01T00:00:00.000Z",  
        "age": 14  
      },  
      "phone": "212-555-1234",  
      "cell": "917-555-5678",  
      "id": {  
        "name": "SSN",  
        "value": "123-45-6789"  
      },  
      "picture": {  
        "large": "https://randomuser.me/api/portraits/men/47.jpg",  
        "medium": "https://randomuser.me/api/portraits/med/men/47.jpg",  
        "thumbnail": "https://randomuser.me/api/portraits/thumb/men/47.jpg"  
      },  
      "nat": "US"  
    }  
  ]  
}
```

Step 6: Now let's fetch this data using HTTP, create a function *fetchUsers* , and call the get function by HTTP instances. It returns the list of objects.

```
Future<List<dynamic>> fetchUsers() async {
  var response =
    await http.get(Uri.parse("https://randomuser.me/api/?results=20"));
  return jsonDecode(response.body) ['results'];
}
```

Step 7: We have to call the *fetchUsers()*. But it is good to load data at initializing our APP, for that, we have a *init()* method.

Ohh great, we fetch the data successfully, now It's time to display it.

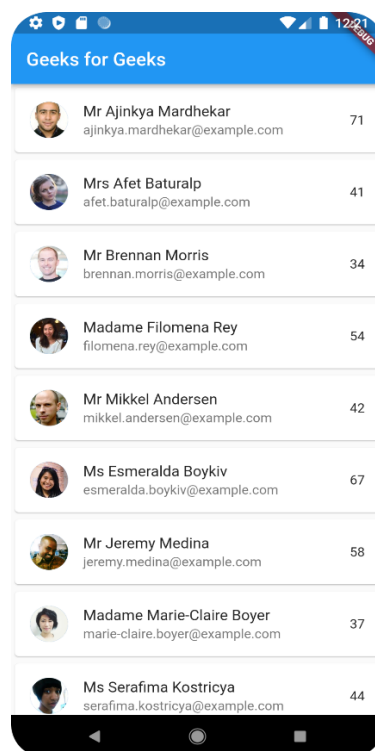
```
@override
void initState() {
  response = fetchUsers();
  super.initState();
}
```

Step 8: Create a *myApiWidget()*.

```
FutureBuilder<dynamic> myApiWidget() {
  return FutureBuilder(
    future: response,
    builder: (BuildContext context, AsyncSnapshot snapshot) {
      if (snapshot.hasData) {
        return ListView.builder(
          itemCount: snapshot.data.length,
          itemBuilder: (BuildContext context, int index) {
            return Card(
              child: Column(
                children: [
                  ListTile(
                    title: Text(snapshot.data[index] ['name'] ['title'] +
                      " " +
                      snapshot.data[index] ['name'] ['first'] +
                      " " +
                      snapshot.data[index] ['name'] ['last']),
                    trailing:
                      Text(snapshot.data[index] ['dob'] ['age'].toString()),
                    leading: CircleAvatar(
                      backgroundImage: NetworkImage(
                        snapshot.data[index] ['picture'] ['large'],
                      ),
                    ),
                  ),
                ],
              ),
            ),
          ),
        );
      }
    },
  );
}
```

```
        subtitle: Text(snapshot.data[index]['email']),  
      ),  
    ],  
  ),  
);  
},  
);  
} else {  
  return Center(  
    child: CircularProgressIndicator(),  
  );  
}  
},  
);  
}
```

Output:



Flutter CRUD Application using PHP REST API



As a Flutter beginner, We all know that We should learn about fundamental usages which applying frequently in a mobile application. So far, We have done tutorials of those essentials. This is another step to create an API using PHP and MySQL. I hope this will really helpful. With this one, you can learn so much of what you do not know. I have written about what are the CRUD operations and how they implement with SQLite and Firebase. This story will describe how to perform CRUD operations with PHP and MySQL. I repeat that this is a very useful application.

What are the CRUD operations?

The basic four operations which we can perform with data. They are,

1. **Create**
2. **Read**
3. **Update**
4. **Delete**
5. **Select**

Pre-requisites

First you have to start Xampp service and start apache service and MySQL service if you are using Windows OR if you are using MacOS you can use MySQL workbench

Explanation

First you have to start Xampp service and start apache service and MySQL service if you are using Windows OR if you are using MacOS you can use MySQL workbench

Package

Go to pubspec.yaml file and add http package under dependencies. (You can find the package [here](#).)

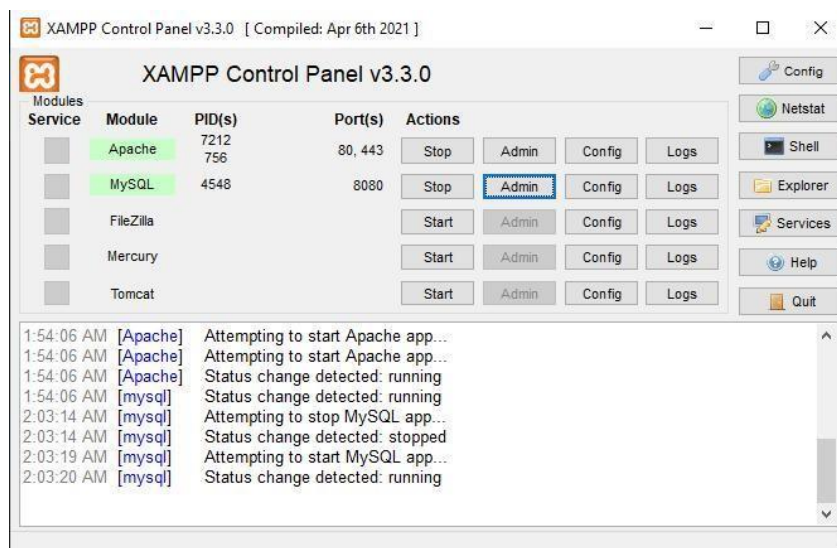
```
dependencies:  
  http: <latest_version>
```

You can **import** the http package as:

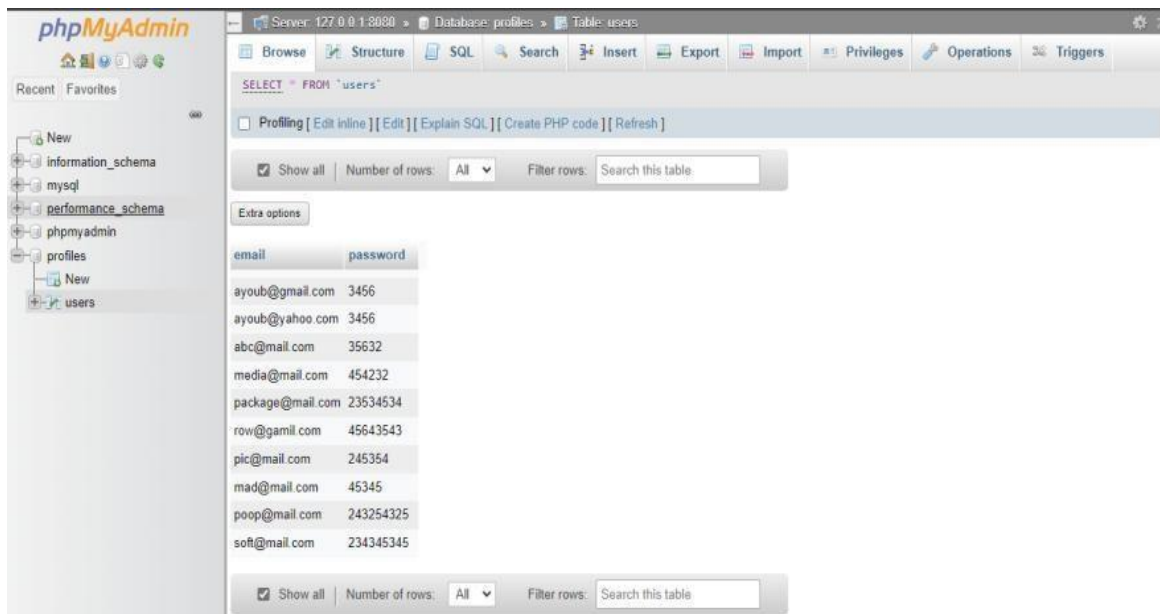
```
import 'package:http/http.dart' as http;
```

Step1: After install any apache + PHP + MySQL stack (XAMPP/ WAMP)

First thing you have to do is to create a database inside MySQL. If you are using Windows, use Xampp controller and start Apache Server and also MySQL server.



You can edit the port in config option or leave it default and click on Admin option to lead the phpMyAdmin



As you can see that, it's running on local host. You can click on *New option to create a database such as I created profiles, and its table is users and as you see the data inside that I inserted using flutter.

You can also create table using packages mentioned above, but I prefer it creating inside phpMyAdmin. The *new option inside the profiles is to create a table.

Database Implementation

First, we want to set up the database and the table. we are going to implement a Inventory Product app. Create a database call **my_store** and two tables name **tb_item** and **users**. The two table structure as follows.

my_store DataBase



tb_item table

Table: tb_item « Database: my_store « 127.0.0.1 الخادم										
استعراض	بناء	SQL	إبحث	إدخال	تصدير	استيراد	الإمميزات	عمليات	تتبع	القوالب
Relation view   بنية الجدول										
#	الاسم	النوع	التجميع	الخواص	خالي	إفتراضي	التعليقات	إضافي	العملية	
1	id	int(11)			لا	لا شيء		AUTO_INCREMENT	تغيير  سقّط  أكثر	☐
2	item_code	text	latin1_swedish_ci		لا	لا شيء			تغيير  سقّط  أكثر	☐
3	item_name	text	latin1_swedish_ci		لا	لا شيء			تغيير  سقّط  أكثر	☐
4	price	int(11)			لا	لا شيء			تغيير  سقّط  أكثر	☐
5	stock	int(11)			لا	لا شيء			تغيير  سقّط  أكثر	☐

users table

Table: users « Database: my_store « 127.0.0.1 الخادم										
استعراض	بناء	SQL	إبحث	إدخال	تصدير	استيراد	الإمميزات	عمليات	تتبع	
Relation view   بنية الجدول										
#	الاسم	النوع	التجميع	الخواص	خالي	إفتراضي	التعليقات	إضافي	العملية	
1	UID	int(100)			لا	لا شيء			تغيير  سقّط  أكثر	☐
2	NAME	text	latin1_swedish_ci		لا	لا شيء			تغيير  سقّط  أكثر	☐
3	UNAME	text	latin1_swedish_ci		لا	لا شيء			تغيير  سقّط  أكثر	☐
4	UPASS	int(10)			لا	لا شيء			تغيير  سقّط  أكثر	☐

PHP Implementation

Before every action with PHP, we want to keep connect the connection between the Database and the API. The following **conn.php** file has to connect the API with the Database.

conn.php

```
<?php

$connect = new mysqli("localhost","root","","my_store");

if($connect){

}else{
    echo "Connection Failed";
    exit();
}
```

If you are working with XAMPP, this will need to place inside the **htdocs/ my_store** directory.

Then create the PHP files for each and every action (login, create, delete, list and update).

getdata.php

To Get All Items in the Table (getdata.php)

```
<?php
include 'conn.php';

$queryResult=$connect->query("SELECT * FROM tb_item");

$result=array();

while($fetchData=$queryResult->fetch_assoc()){
    $result[]=$fetchData;
}

echo json_encode($result);

?>
```

user_login.php

To Get one specific Item in the Table (user_login.php)

```
<?php

//MySQL database Connection
include 'conn.php';


//Received JSON into $json variable
$json = file_get_contents('php://input');
```

```
//Decoding the received JSON and store into $obj variable.
$obj = json_decode($json,true);

if(isset($obj["username"]) && isset($obj["password"])){

    $uname = mysqli_real_escape_string($connect,$obj['username']);
    $pwd = mysqli_real_escape_string($connect,$obj['password']);

    //Declare array variable
    $result=[];

    //Select Query
    $sql="SELECT * FROM users WHERE UNAME='{ $uname}' and UPASS='{ $pwd}'";
    $res=$connect->query($sql);

    if($res->num_rows>0){

        $row=$res->fetch_assoc();

        $result['loginStatus']=true;
        $result['message']="Login Successfully";

        $result["userInfo"]=$row;

    }else{

        $result['loginStatus']=false;
        $result['message']="Invalid Login Details";
    }

    // Converting the array into JSON format.
    $json_data=json_encode($result);

    // Echo the $json.
    echo $json_data;
}
?>
```

To insert a new student record (adddata.php)

```
<?php

include 'conn.php';

$itemcode = $_POST['itemcode'];
$itemname = $_POST['itemname'];
$price = $_POST['price'];
$stock= $_POST['stock'];
```

```

        $connect->query("INSERT INTO tb_item
(item_code,item_name,price,stock) VALUES
('".$itemcode."','".$itemname."','".$price."','".$stock."')")

?>

```

deleteData.php

To Delete s record from the table (deleteData.php)

```

<?php
    include 'conn.php';
    $id=$_POST['id'];
    $connect->query("DELETE FROM tb_item WHERE id=".$id);

?>

```

editdata.php

To update a record (editdata.php)

```

<?php

    include 'conn.php';

    $id = $_POST['id'];
    $itemcode = $_POST['itemcode'];
    $itemname = $_POST['itemname'];
    $price = $_POST['price'];
    $stock= $_POST['stock'];

    $connect->query("UPDATE tb_item SET item_code='".$itemcode."',
item_name='".$itemname."', price='".$price."', stock='".$stock."' WHERE
id=". $id);

?>

```

Flutter Implementation

Then we want to implement the Flutter code. So, create a new Flutter application as usual. Here I want to emphasize more about the Flutter, I am not going to explain PHP and MySQL more. Continue the following steps one by one. Here you want to focus on how to call to routes and how to handle those results in Flutter. Otherwise, you can create these function for your own purpose.

Inside the **main.dart** file, we only introduce the flowing lines to make you to naviga to Splash Screen:

```
import 'package:flutter/material.dart';

import 'splash_roq.dart';

void main() => runApp(Main_CURD_Sample1_Roq());

class Main_CURD_Sample1_Roq extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: "Inventory Product ",
      // theme: ThemeData(
      //   primarySwatch: Colors.indigo,
      // ),
      home: SplashScreen_Roq(),
    );
  }
}
```

splash_roq.dart

The first loading page is the **splash_roq.dart** page. Inside the splash_roq.dart.

```
import 'dart:async';
import 'package:flutter/material.dart';

import 'Login_roq.dart';

class SplashScreen_Roq extends StatefulWidget {
  @override
  SplashScreenState createState() => new SplashScreenState();

  const SplashScreen_Roq({super.key});
}

class SplashScreenState extends State<SplashScreen_Roq>
  with SingleTickerProviderStateMixin {
  var _visible = true;

  late AnimationController animationController;
  late Animation<double> animation;

  startTime() async {
    var _duration = new Duration(seconds: 10);
    return new Timer(_duration, navigationPage);
  }

  void navigationPage() {
    Navigator.of(context).pushReplacement(MaterialPageRoute(
      builder: (context) =>
```

```

    //LoginForm_Simple(),
    LoginForm_Roq(),
  ));
}

@override
void initState() {
  super.initState();
  animationController = new AnimationController(
    vsync: this, duration: new Duration(seconds: 5));
  animation =
    new CurvedAnimation(parent: animationController, curve: Curves.easeOut);

  animation.addListener(() => this.setState(() {}));
  animationController.forward();

  setState(() {
    _visible = !_visible;
  });
  startTime();
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    body: Container(
      //=> color: Colors.white,
      child: Stack(
        fit: StackFit.expand,
        children: <Widget>[
          Padding(
            padding: EdgeInsets.only(bottom: 180.0),
            child: new Column(
              mainAxisAlignment: MainAxisAlignment.end,
              mainAxisAlignment: MainAxisAlignment.min,
              children: <Widget>[
                Text(
                  "Inventory Product Roq App",
                  style: TextStyle(
                    fontSize: 16.0,
                    color: Colors.black,
                    fontFamily: "Netflix",
                  ),
                  textAlign: TextAlign.center,
                ),
                Text(
                  "ruqsal3@gmail.com",
                  style: TextStyle(
                    fontSize: 14.0,
                    color: Colors.black,
                    fontFamily: "Netflix",
                  ),
                  textAlign: TextAlign.center,
                ),
                Container(height: 20),
                Text(" Version 1.0.0", style: TextStyle(
                  fontStyle: FontStyle.italic,

```

Login_roq.dart

Lecture10: JSON, and Rest API
rugsal3.flutter3@gmail.com

```

Future userLogin() async {
  //Login API URL
  //use your local IP address instead of localhost or use Web API
  String url = "http://10.0.2.2/my_store/user_login.php";
  // Showing LinearProgressIndicator.
  setState(() {
    _visible = true;
  });

  // Getting username and password from Controller
  Map<String, String> data = {
    'username': userController.text,
    'password': pwdController.text,
  };

  //Starting Web API Call.
  http.Response response =
    await http.post(Uri.parse(url), body: json.encode(data));
  if (response.statusCode == 200) {
    //Server response into variable
    print("The User Data is : ${response.body}");
    var msg = jsonDecode(response.body);

    //Check Login Status
    if (msg['loginStatus'] == true) {
      setState(() {
        //hide progress indicator
        _visible = false;
      });

      // Navigate to Home Screen
      Navigator.push(
        context,
        MaterialPageRoute(
          builder: (context) => HomePage_Roq(msg['userInfo']['NAME'])));
    } else {
      setState(() {
        //hide progress indicator
        _visible = false;

        //Show Error Message Dialog
        showMessage(msg["message"]);
      });
    }
  } else {
    setState(() {
      //hide progress indicator
      _visible = false;

      //Show Error Message Dialog
      showMessage("Error during connecting to Server.");
    });
  }
}

Future<dynamic> showMessage(String _msg) async {
  showDialog(

```

```

context: context,
builder: (BuildContext context) {
  return Dialog(
    backgroundColor: Colors.transparent,
    child: Container(
      width: 160,
      child: Card(
        shape: RoundedRectangleBorder(
          borderRadius: BorderRadius.circular(4),
        ),
        color: Colors.white,
        clipBehavior: Clip.antiAliasWithSaveLayer,
        child: Wrap(
          children: <Widget>[
            Container(
              padding: EdgeInsets.all(20),
              width: double.infinity,
              color: Color(0xffe46b10),
              child: Column(
                children: <Widget>[
                  Container(height: 10),
                  Icon(Icons.cloud_off, color: Colors.white, size: 80),
                  Container(height: 10),
                  Text("Error !",
                    style: Theme.of(context)
                      .textTheme
                      .headline6
                      ?.copyWith(color: Colors.white)),
                  Container(height: 10),
                ],
              ),
            ),
            Container(
              padding: EdgeInsets.all(20),
              width: double.infinity,
              child: Column(
                children: <Widget>[
                  Text("$ _msg",
                    textAlign: TextAlign.center,
                    style: Theme.of(context)
                      .textTheme
                      .subtitle1
                      ?.copyWith(color: Color(0xFF666666))),
                  Container(height: 10),
                  TextButton(
                    style: TextButton.styleFrom(
                      padding: EdgeInsets.symmetric(
                        vertical: 0, horizontal: 40),
                      shape: RoundedRectangleBorder(
                        borderRadius: new BorderRadius.circular(18.0)),
                      backgroundColor: Color(0xffe46b10),
                    ),
                    child: Text(
                      "Retry",
                      style: TextStyle(color: Colors.white),
                    ),
                    onPressed: () {

```

```

        Navigator.of(context).pop();
      },
    ),
    1,
  ),
),
1,
),
),
),
);
/* AlertDialog(

  title: new Text(_msg),
  actions: <Widget>[
    TextButton(
      child: new Text("OK"),
      onPressed: () {
        Navigator.of(context).pop();
      },
    ),
  ],
);*/
},
);
}

//////////////////////////////////////
void error(BuildContext context, String error) {
  final scaffold = ScaffoldMessenger.of(context);
  scaffold.showSnackBar(
    SnackBar(
      content: Text(error),
      action: SnackBarAction(
        label: 'OK', onPressed: scaffold.hideCurrentSnackBar),
    ),
  );
}

bool password_visiable = false;

Widget _divider() {
  return Container(
    margin: EdgeInsets.symmetric(vertical: 10),
    child: Row(
      children: <Widget>[
        SizedBox(
          width: 20,
        ),
        Expanded(
          child: Padding(
            padding: EdgeInsets.symmetric(horizontal: 10),
            child: Divider(
              thickness: 1,
            ),
          ),
        ),
      ],
    ),
    Text('or'),
  );
}

```



```

    ),
  ),
],
),
);
}

Widget _createAccountLabel() {
  return InkWell(
    onTap: () {
      Navigator.push(
        context, MaterialPageRoute(builder: (context) => SignUpPage_Roq());
      ),
    child: Container(
      margin: EdgeInsets.symmetric(vertical: 20),
      padding: EdgeInsets.all(15),
      alignment: Alignment.bottomCenter,
      child: Row(
        mainAxisAlignment: MainAxisAlignment.center,
        children: <Widget>[
          Text(
            'Don\'t have an account ?',
            style: TextStyle(fontSize: 13, fontWeight: FontWeight.w600),
          ),
          SizedBox(
            width: 10,
          ),
          Text(
            'Register',
            style: TextStyle(
              color: Color(0xffe46b10),
              fontSize: 13,
              fontWeight: FontWeight.w600),
          ),
        ],
      ),
    ),
  );
}

Widget _title() {
  return RichText(
    textAlign: TextAlign.center,
    text: TextSpan(
      text: 'L',
      style: TextStyle(
        fontSize: 30,
        fontWeight: FontWeight.w700,
        color: Color(0xffe46b10)),
      children: [
        TextSpan(
          text: 'og',
          style: TextStyle(color: Colors.black, fontSize: 30),
        ),
        TextSpan(
          text: 'in',
          style: TextStyle(color: Color(0xffe46b10), fontSize: 30),

```

```

    ),
  1),
);
}

@override
Widget build(BuildContext context) {
  final height = MediaQuery.of(context).size.height;
  return new Scaffold(
    resizeToAvoidBottomInset: true,
    body: Padding(
      padding: const EdgeInsets.all(20.0),
      child: ListView(
        children: <Widget>[
          new Column(
            children: <Widget>[
              SizedBox(
                height: 10,
              ),
              Image(image: AssetImage("asset_roq/images/login.png")),
              //SizedBox(height: height * .2),
              _title(),

              Form(
                key: _formKey,
                child: Padding(
                  padding: const EdgeInsets.all(8.0),
                  child: Column(
                    children: [
                      Theme(
                        data: new ThemeData(
                          primaryColor: Color.fromRGBO(84, 87, 90, 0.5),
                          primaryColorDark: Color.fromRGBO(84, 87, 90, 0.5),
                          hintColor: Color.fromRGBO(
                            84, 87, 90, 0.5), //placeholder color
                        ),
                      child: TextFormField(
                        controller: userController,
                        decoration: InputDecoration(
                          focusedBorder: new OutlineInputBorder(
                            borderSide: BorderSide(
                              color: Color.fromRGBO(84, 87, 90, 0.5),
                              style: BorderStyle.solid,
                            ),
                          ),
                          enabledBorder: new OutlineInputBorder(
                            borderSide: BorderSide(
                              color: Color.fromRGBO(84, 87, 90, 0.5),
                              style: BorderStyle.solid,
                            ),
                          ),
                          errorBorder: new OutlineInputBorder(
                            borderSide: BorderSide(
                              color: Colors.red,
                              width: 1.0,
                              style: BorderStyle.solid,
                            ),
                          ),

```

[illegible]

```

        errorBorder: new OutlineInputBorder(
          borderSide: BorderSide(
            color: Colors.red,
            width: 1.0,
            style: BorderStyle.none,
          ),
        ),
        border: new OutlineInputBorder(
          borderSide: BorderSide(
            color: Color.fromRGBO(84, 87, 90, 0.5),
            style: BorderStyle.none,
          ),
        ),
        labelText: 'Enter Password',
        prefixIcon: const Icon(
          Icons.lock,
          color: Color.fromRGBO(84, 87, 90, 0.5),
        ),
        hintText: 'Password',
      ),
      keyboardType: TextInputType.visiblePassword,
      obscureText: !password_visiable,
      validator: (value) {
        if (value == null || value.isEmpty) {
          return 'Please Enter Password';
        }
        return null;
      },
    ),
    SizedBox(
      height: 20.0,
    ),
    Padding(
      padding: const EdgeInsets.all(8.0),
      child: ElevatedButton(
        onPressed: () => {
          // Validate returns true if the form is valid, or false
          otherwise.

          if (_formKey.currentState!.validate())
            {userLogin()}
        },
        child: Padding(
          padding: EdgeInsets.all(16.0),
          child: Text(
            'Login',
            style: TextStyle(fontSize: 18.0),
          ),
        ),
        style: ButtonStyle(
          backgroundColor: MaterialStateProperty.all<Color>(
            Theme.of(context).primaryColor,
          ),
        ),
      ),
    ),
    Container(
      padding: EdgeInsets.symmetric(vertical: 10),

```

homepage_roq.dart

Lecture10: JSON, and Rest API
rugsal3.flutter3@gmail.com

```

        style: TextStyle(
          fontFamily: "Netflix",
          fontSize: 18,
          color: Colors.pink,
          fontWeight: FontWeight.bold),
      ),
    ),
    SizedBox(
      height: 20,
    ),
    Image(image: AssetImage("asset_roq/images/dashboard.png")),
    SizedBox(
      height: 40,
    ),
    Container(
      height: 50,
      width: double.infinity,
      child: ElevatedButton(
        style: ElevatedButton.styleFrom(backgroundColor: Colors.pink),
        child: Text(
          "View Data",
          style: TextStyle(fontFamily: "Netflix", color: Colors.white),
        ),
        onPressed: () {
          Navigator.push(context, MaterialPageRoute(builder: (context) {
            return Get_ListDataInventory_Roq();
          }));
        }
      ),
    ),
    SizedBox(
      height: 20,
    ),
    Container(
      height: 50,
      width: double.infinity,
      child: ElevatedButton(
        //color: Colors.blue,
        child: Text(
          "Logout",
          style: TextStyle(fontFamily: "Netflix", color: Colors.white),
        ),
        onPressed: () {
          Navigator.pushReplacement(context,
            MaterialPageRoute(builder: (context) {
              return LoginForm_Roq();
            }));
          logout(context);
        }
      ),
    ),
  ),
),
),
));
}
}

void logout(BuildContext context) {
  final scaffold = ScaffoldMessenger.of(context);

```

```
scaffold.showSnackBar(
  SnackBar(
    content: Text("You have successfully logged out!"),
    action:
      SnackBarAction(label: 'OK', onPressed: scaffold.hideCurrentSnackBar),
  ),
);
}
```

listdata_roq.dart

Inside the Get_ListDataInventory_Roq page, we create a list view to extract all list of items. and list them in a list view. They have a floating button to create a new record and each list item navigate to its detail view then tapping on that item.

```
import 'dart:async';
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

import 'adddata_roq.dart';
import 'detail_roq.dart';

import 'homepage_roq.dart';

class Get_ListDataInventory_Roq extends StatefulWidget {
  @override
  // final String text;
  // Home({Key key, @required this.text}) : super(key: key);
  _Get_ListDataInventory_RoqState createState() => new
  _Get_ListDataInventory_RoqState();
}

class _Get_ListDataInventory_RoqState extends State<Get_ListDataInventory_Roq> {
  @override
  void initState() {
    super.initState();
    getData();
  }

  Future<List> getData() async {
    final http.Response response = await
http.get(Uri.parse("http://10.0.2.2/my_store/getdata.php"));
    return json.decode(response.body);
  }

  bool Value = false ;
  @override
  Widget build(BuildContext context) {
    return new Scaffold(
      appBar: new AppBar(
        title: new Text(
          "List Data Inventory",
          style: TextStyle(fontFamily: "Netflix"),

```

```

),
// backgroundColor: Colors.pink,
actions: [
  PopupMenuButton(
    // color: Colors.yellowAccent,
    elevation: 20,
    enabled: true,
    onSelected: (Value){

      if (Value)
      {
        Navigator.pushReplacement(context,
          MaterialPageRoute(builder: (context) {
            return AddData_Roq();
          }
        )
      );
      }
      else
      {
        return null;
      }
    }

    ,
    itemBuilder: (context) => [
      PopupMenuItem(
        child: Text("Privacy"),
      ),
      PopupMenuItem(
        child: Text("Settings"),
      ),
      PopupMenuItem(
        child: Text("About us"),
        value: !Value,
      )
    ]
  )
],
),
floatingActionButton: Row(
  mainAxisAlignment: MainAxisAlignment.end,
  children: <Widget>[
    FloatingActionButton.extended(
      heroTag: "logout",
      onPressed: () {
        Navigator.of(context).push(new MaterialPageRoute(
          builder: (BuildContext context) => new HomePage_Roq()));
        // logout(context);
      },
      // onPressed: () => Navigator.of(context).pushReplacement(
      //   new MaterialPageRoute(
      //     builder: (BuildContext context) => new AddData()),
      label: const Text('Menu'),
      icon: const Icon(Icons.logout),
      backgroundColor: Colors.blue,
    ),
    SizedBox(

```

```

        width: 20,
      ),
      //SizedBox(height: 10,),
      FloatingActionButton.extended(
        heroTag: "add data",
        onPressed: () {
          Navigator.of(context).push(new MaterialPageRoute(
            builder: (BuildContext context) => new AddData_Roq()));
        },
        label: const Text('Add Data'),
        icon: const Icon(Icons.add_box),
        backgroundColor: Colors.pink,
      ),
    ],
  ),
),

body:
  Container(
    //    color: Colors.purple,
    height: double.infinity,
    child: new FutureBuilder(
      future: getData(),
      builder: (context, snapshot) {
        if (snapshot.hasError)
          print(snapshot.error);

        return snapshot.hasData
          ? ItemList(
              listItems_roq: snapshot.data!,
            )
          : new Center(
              child: new CircularProgressIndicator(),
            );
      },
    ),
  ),
);
}
}

class ItemList extends StatelessWidget {
  final List listItems_roq;
  ItemList({required this.listItems_roq});

  @override
  Widget build(BuildContext context) {
    return new ListView.builder(
      itemCount: listItems_roq == null ? 0 : listItems_roq.length,
      itemBuilder: (context, i) {
        return new Container(

          padding: const EdgeInsets.all(0.0),
          child: new GestureDetector(
            onTap: () =>
              Navigator.of(context).push(new MaterialPageRoute(

```

```

        builder: (BuildContext context) => new Detail_Roq(
          list: listItems_roq,
          index: i,
        )),
      child: new Card(
        elevation: 10,
        child: new ListTile(
          trailing: Icon(Icons.description),
          title: new Text(
            listItems_roq[i]['item_name'],
            style: TextStyle(fontFamily: "Netflix"),
          ),
          leading: new Icon(Icons.widgets),
          subtitle: new Text("Stock : ${listItems_roq[i]['stock']}"),
        ),
      ),
    ),
  );
},
);
}
}

```

adddata_roq.dart

The following page is used to create and add a new item record to the **tb_item** table.

```

import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

import 'listdata_roq.dart';

class AddData_Roq extends StatefulWidget {
  @override
  _AddDataState createState() => new _AddDataState();
}

class _AddDataState extends State<AddData_Roq> {
  TextEditingController controllerCode = new TextEditingController();
  TextEditingController controllerName = new TextEditingController();
  TextEditingController controllerPrice = new TextEditingController();
  TextEditingController controllerStock = new TextEditingController();

  void addData() {
    var url = "http://10.0.2.2/my_store/adddata.php";
    http.post(Uri.parse(url), body: {
      "itemcode": controllerCode.text,
      "itemname": controllerName.text,
      "price": controllerPrice.text,
      "stock": controllerStock.text
    });
  }
}

```

```

void error(BuildContext context, String error) {
  final scaffold = ScaffoldMessenger.of(context);
  scaffold.showSnackBar(
    SnackBar(
      content: Text(error),
      action: SnackBarAction(
        label: 'OK', onPressed: scaffold.hideCurrentSnackBar),
    ),
  );
}

@override
Widget build(BuildContext context) {
  return new Scaffold(
    appBar: new AppBar(
      title: new Text("Add Data", style: TextStyle(fontFamily: "Netflix")),
    ),
    body: Padding(
      padding: const EdgeInsets.all(20.0),
      child: ListView(
        children: <Widget>[
          new Column(
            children: <Widget>[
              new TextField(
                controller: controllerCode,
                style: TextStyle(fontFamily: "Netflix", fontSize: 15),
                decoration: new InputDecoration(
                  hintText: "Item Code",
                  hintStyle: TextStyle(fontFamily: "Netflix"),
                  labelText: "Item Code",
                  labelStyle: TextStyle(fontFamily: "Netflix")),
              ),
              new TextField(
                controller: controllerName,
                style: TextStyle(fontFamily: "Netflix", fontSize: 15),
                decoration: new InputDecoration(
                  hintText: "Item Name",
                  hintStyle: TextStyle(fontFamily: "Netflix"),
                  labelText: "Item Name"),
              ),
              new TextField(
                controller: controllerPrice,
                style: TextStyle(fontFamily: "Netflix", fontSize: 15),
                decoration: new InputDecoration(
                  hintText: "Price", labelText: "Price"),
              ),
              new TextField(
                controller: controllerStock,
                style: TextStyle(fontFamily: "Netflix", fontSize: 15),
                decoration: new InputDecoration(
                  hintText: "Stock", labelText: "Stock"),
              ),
              new Padding(
                padding: const EdgeInsets.all(10.0),
              ),
              new Column(
                children: <Widget>[

```



```

    }
  }
class showMessage extends StatelessWidget {
  const showMessage({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return Container(
      child: AlertDialog(
        title: Text("Data has been added secssfully"),
        actions: [
          TextButton(
            onPressed: () {
              Navigator.of(context).pushReplacement(new MaterialPageRoute (
                builder: (BuildContext context) => new
Get_ListDataInventory_Roq()));
              // Navigator.of(context).pop();

              //Get_ListDataInventory_Roq();
            },
            child: Text("OK"))
        ],
      ),
    );
  }
}

```

detail_roq.dart

The details page is showing the currently selected item, User can also delete or, edit it. The current Item received was the response to the HTTP post request.

```

import 'dart:ffi';

import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
import 'editdata_roq.dart';

import 'listdata_roq.dart';

class Detail_Roq extends StatefulWidget {
  List list;
  int index;

  Detail_Roq({required this.index, required this.list});

  @override
  _Detail_RoqState createState() => new _Detail_RoqState();
}

class _Detail_RoqState extends State<Detail_Roq> {
  void deleteData() {
    var url = "http://10.0.2.2/my_store/deleteData.php";
    http.post(Uri.parse(url), body: {'id': widget.list[widget.index]['id']});
    Navigator.pushReplacement(context, MaterialPageRoute(builder: (context) {
      return Get_ListDataInventory_Roq();
    }));
  }
}

```

```

    ));
    _DeleteData(context, "Data deleted successfully");
}

void _DeleteData(BuildContext context, String error) {
  final scaffold = ScaffoldMessenger.of(context);
  scaffold.showSnackBar(
    SnackBar(
      content: Text(error),
      action: SnackBarAction(
        label: 'OK', onPressed: scaffold.hideCurrentSnackBar),
    ),
  );
}

@override
Widget build(BuildContext context) {
  return new Scaffold(
    appBar: new AppBar(
      title: new Text(
        "${widget.list[widget.index]['item_name']}",
        style: TextStyle(fontFamily: "Netflix"),
      ),
    ),
    body: new Container(
      height: 370.0,
      padding: const EdgeInsets.all(20.0),
      child: new Card(
        child: new Center(
          child: new Column(
            children: <Widget>[
              new Padding(
                padding: const EdgeInsets.only(top: 30.0),
              ),
              new Text(
                widget.list[widget.index]['item_name'],
                style: new TextStyle(fontFamily: "Netflix", fontSize: 20.0),
              ),
              new Text(
                "Code : ${widget.list[widget.index]['item_code']}",
                style: new TextStyle(fontSize: 18.0),
              ),
              new Text(
                "Price : ${widget.list[widget.index]['price']}.RY",
                style: new TextStyle(fontSize: 18.0),
              ),
              new Text(
                "Stock : ${widget.list[widget.index]['stock']}",
                style: new TextStyle(fontSize: 18.0),
              ),
              // new Padding(
              //   padding: const EdgeInsets.only(top: 30.0),
              // ),
              new Column(
                //mainAxisSize: MainAxisSize.min,
                children: <Widget>[
                  Container(

```

```

width: double.infinity,
padding: EdgeInsets.fromLTRB(10, 5, 10, 5),
// Here An Editing !!!!!!!!!!!!!!!!!!!!!!!
child: new OutlinedButton(
  child: new Text(
    "EDIT",
    style: TextStyle(
      fontFamily: "Netflix", color: Colors.white),
  ),
  style: OutlinedButton.styleFrom(
    backgroundColor: Colors.green),
  onPressed: () => Navigator.of(context)
    .pushReplacement(new MaterialPageRoute(
      builder: (BuildContext context) => new EditData_Roq(
        list: widget.list,
        index: widget.index,
      ),
    )),
),
),
Container(
  width: double.infinity,
  padding: EdgeInsets.fromLTRB(10, 5, 10, 5),
  child: new TextButton(
    child: new Text(
      "DELETE",
      style: TextStyle(
        fontFamily: "Netflix", color: Colors.white),
    ),
    style: ElevatedButton.styleFrom(
      shape: RoundedRectangleBorder(
        //Editing !!!!!!!!!!!!!!!!!!!!!!!
        borderRadius: BorderRadius.circular(35),
      ),
      backgroundColor: Colors.red),
    onPressed: () => deleteData(),
  ),
),
1,
),
// new Padding(
//   padding: const EdgeInsets.only(top: 30.0),
// ),
Container(
  padding: EdgeInsets.fromLTRB(10, 5, 10, 5),
  width: double.infinity,
  child: ElevatedButton(
    child: Text(
      "Go Home",
      style: TextStyle(color: Colors.white),
    ),
    style:
      ElevatedButton.styleFrom(backgroundColor: Colors.blue),
    onPressed: () {
      Navigator.pushReplacement(context,
        MaterialPageRoute(builder: (context) {
          return Get_ListDataInventory_Roq();
        }

```

```

    ));
  },
),
1,
),
),
),
),
);
}
}

```

editdata_roq.dart

The following code snippet is the **editdata_roq.dart** page. User can edit the particular record using a POST request.

```

import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
import 'listdata_roq.dart';

class EditData_Roq extends StatefulWidget {
  final List list;
  final int index;

  EditData_Roq({required this.list, required this.index});

  @override
  _EditData_RoqState createState() => new _EditData_RoqState();
}

class _EditData_RoqState extends State<EditData_Roq> {
  late TextEditingController controllerCode;
  late TextEditingController controllerName;
  late TextEditingController controllerPrice;
  late TextEditingController controllerStock;

  void editData() {
    var url = "http://10.0.2.2/my_store/editdata.php";
    http.post(Uri.parse(url), body: {
      "id": widget.list[widget.index]['id'],
      "itemcode": controllerCode.text,
      "itemname": controllerName.text,
      "price": controllerPrice.text,
      "stock": controllerStock.text
    }));
  }

  @override
  void initState() {
    controllerCode =
      new TextEditingController(text: widget.list[widget.index]['item_code']);
    controllerName =
      new TextEditingController(text: widget.list[widget.index]['item_name']);
    controllerPrice =

```

```

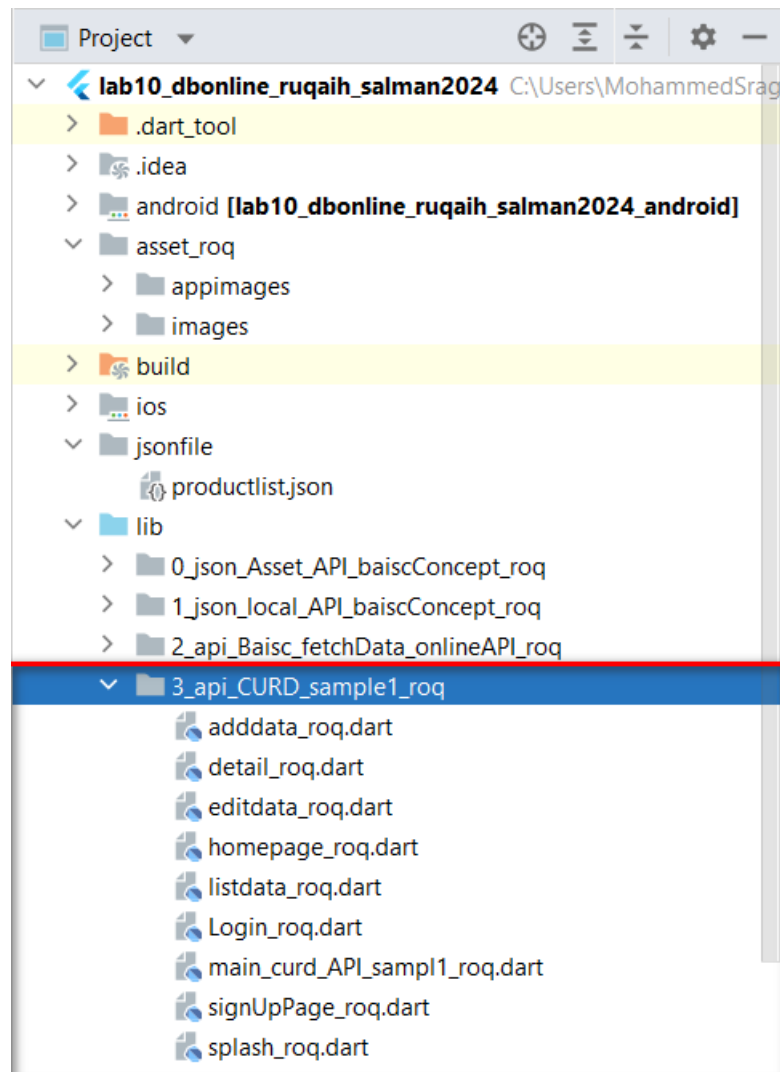
        new TextEditingController(text: widget.list[widget.index]['price']);
        controllerStock =
            new TextEditingController(text: widget.list[widget.index]['stock']);
        super.initState();
    }

    @override
    Widget build(BuildContext context) {
        return new Scaffold(
            appBar: new AppBar(
                title: new Text(
                    "Edit Data",
                    style: TextStyle(fontFamily: "Netflix"),
                ),
            ),
            body: Padding(
                padding: const EdgeInsets.all(20.0),
                child: ListView(
                    children: <Widget>[
                        new Column(
                            children: <Widget>[
                                new TextField(
                                    controller: controllerCode,
                                    style: TextStyle(fontFamily: "Netflix", fontSize: 15),
                                    decoration: new InputDecoration(
                                        hintText: "Item Code", labelText: "Item Code"),
                                ),
                                new TextField(
                                    controller: controllerName,
                                    style: TextStyle(fontFamily: "Netflix", fontSize: 15),
                                    decoration: new InputDecoration(
                                        hintText: "Item Name", labelText: "Item Name"),
                                ),
                                new TextField(
                                    controller: controllerPrice,
                                    style: TextStyle(fontFamily: "Netflix", fontSize: 15),
                                    decoration: new InputDecoration(
                                        hintText: "Price", labelText: "Price"),
                                ),
                                new TextField(
                                    controller: controllerStock,
                                    style: TextStyle(fontFamily: "Netflix", fontSize: 15),
                                    decoration: new InputDecoration(
                                        hintText: "Stock", labelText: "Stock"),
                                ),
                                SizedBox(
                                    height: 30,
                                ),
                                Container(
                                    width: double.infinity,
                                    height: 50,
                                    child: ElevatedButton(
                                        child: new Text(
                                            "EDIT DATA",
                                            style:
                                                TextStyle(fontFamily: "Netflix", color: Colors.white),
                                        ),
                                    ),
                                ),
                            ],
                        ),
                    ],
                ),
            ),
        );
    }

```

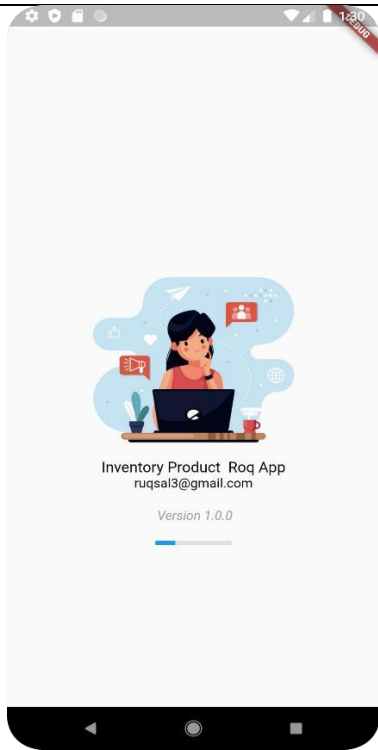
```
Get_ListDataInventory_Roq())));
```

Lecture10: JSON, and Rest API
ruqsal3.flutter3@gmail.com

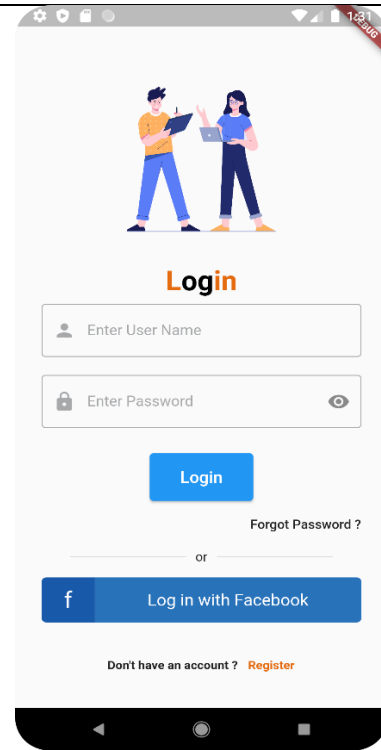


Finally, I have sketched the whole application scenario as follows. you can run and see the output on your emulator.

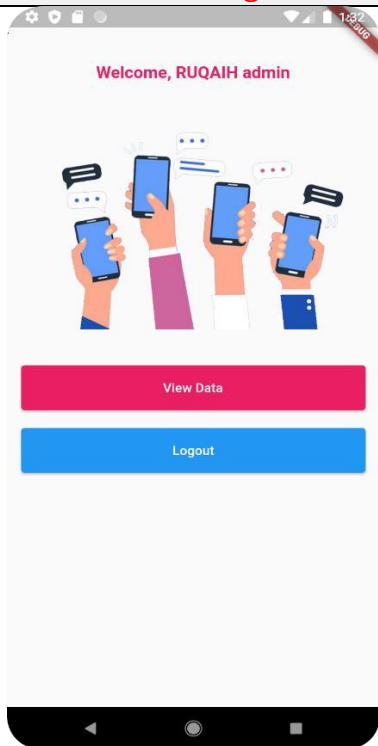
Splash Page	Login Page
--------------------	-------------------



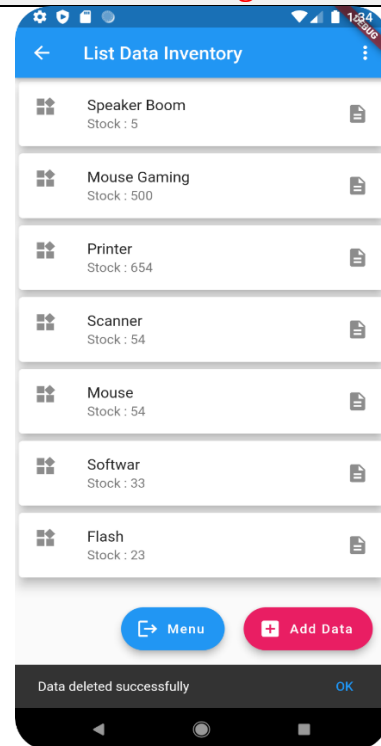
Home Page



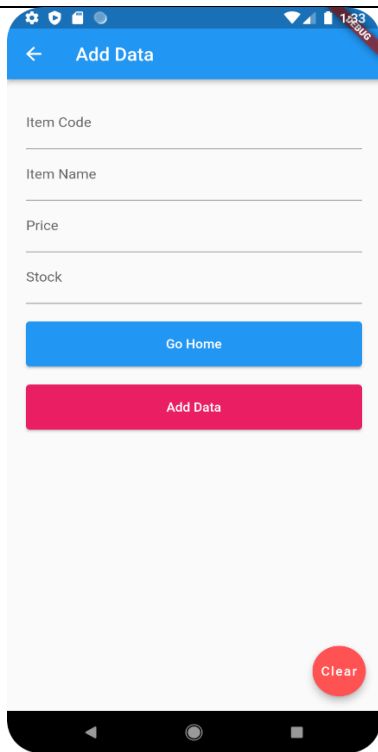
List Page



Add Data Page



Datils Page



← Add Data

Item Code

Item Name

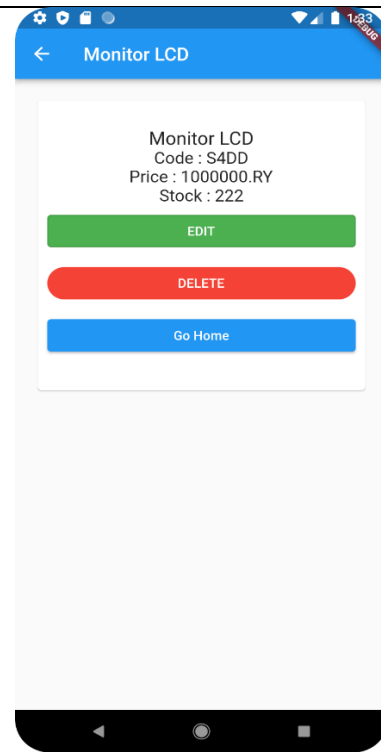
Price

Stock

Go Home

Add Data

Clear



← Monitor LCD

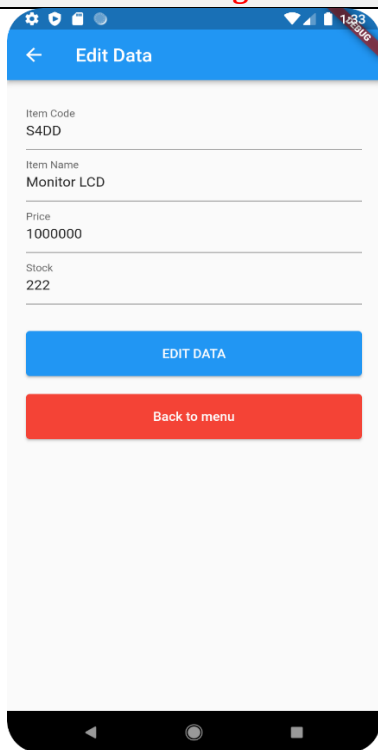
Monitor LCD
Code : S4DD
Price : 1000000.RY
Stock : 222

EDIT

DELETE

Go Home

Edit Page



← Edit Data

Item Code
S4DD

Item Name
Monitor LCD

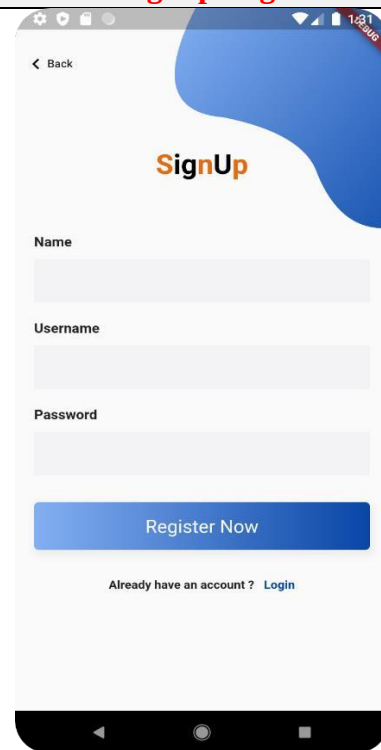
Price
1000000

Stock
222

EDIT DATA

Back to menu

Signup Page



← Back

SignUp

Name

Username

Password

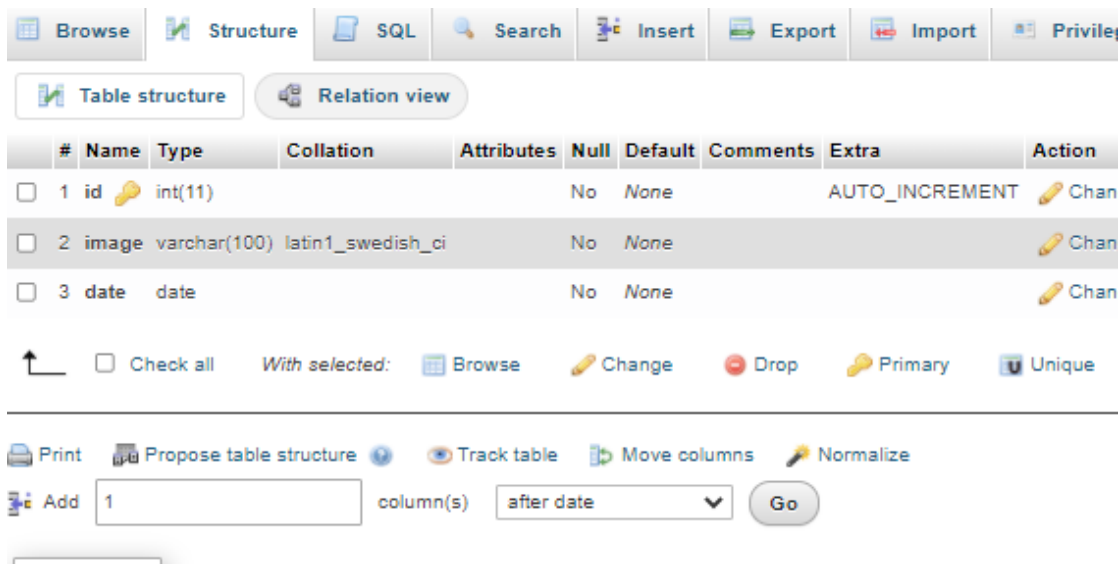
Register Now

Already have an account ? [Login](#)

How to Upload Image in Flutter & Save It in MySQL Database

On this occasion, we will continue and refine the previous discussion, namely creating an image upload feature and saving it into the database. In this tutorial we will learn, http requests, image picker, grid view builder, snackbar, create back-ends, and learn other widget implementations.

Step1: First we will create the back-end part first. So please create a database with the name of **practice** and create a table in it with the name **flutter_upload_images**. Then follow the table structure as shown below.



The screenshot shows the MySQL Table Structure for a table named 'flutter_upload_images'. The table has three columns: 'id' (int(11), primary key, auto-increment), 'image' (varchar(100), latin1_swedish_ci), and 'date' (date). The interface includes tabs for 'Table structure' and 'Relation view', and a toolbar with various actions like 'Check all', 'Browse', 'Change', 'Drop', 'Primary', and 'Unique'.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	int(11)			No	None		AUTO_INCREMENT	Chan
2	image	varchar(100)	latin1_swedish_ci		No	None			Chan
3	date	date			No	None			Chan

Table of flutter_upload_images

Step2: Next, create a new folder in the **htdocs** folder with the name **flutter_upload_image** or with a free name. Then create a new file in it with the name **create.php**. Follow the script as below.

```
<?php

$connection = new mysqli("localhost","root","","practice");

//get image name
$image = $_FILES['image']['name'];

//create date now
$date = date('Y-m-d');

//make image path
$imagePath = 'images/'.$image;

$tmp_name = $_FILES['image']['tmp_name'];
```

```
//move image to images folder
move_uploaded_file($tmp_name, $imagePath);

$result = mysqli_query($connection, "insert into flutter_upload_images set
image='$image', date='$date'");

if($result){
    echo json_encode([
        'message' => 'Data input successfully'
    ]);
}else{
    echo json_encode([
        'message' => 'Data Failed to input'
    ]);
}
?>
```

Step3. Create a new file in the **flutter_upload_image** folder with the name **list.php**. Then follow the script as below.

```
<?php

$connection = new mysqli("localhost","root","","practice");
$data       = mysqli_query($connection, "select * from flutter_upload_images");
$data       = mysqli_fetch_all($data, MYSQLI_ASSOC);

echo json_encode($data);
```

Step4. Don't forget to create an **images** folder inside the **flutter_upload_image** folder. This folder serves to hold all the images later. So that the contents in the **flutter_upload_image** folder are the **create.php** file, the **list.php** file, and the **images** folder.

Step5. Next, create a new flutter project with the name **flutter_upload_image**. And in this tutorial I used the null safety version of Flutter.

Step6. After creating a new flutter project, then open the **pubspec.yaml** file. Then add the package **image_picker:** and package **http:** (no need to add the package version) Don't forget to press CTRL + S or save the file to download the required package. See the image below for the full details, (no need to add a package version).

```
dependencies:
  flutter:
    sdk: flutter

  # The following adds the Cupertino Icons font to your application.
  # Use with the CupertinoIcons class for iOS style icons.
  cupertino_icons: ^1.0.2
  image_picker:
  http:

dev_dependencies:
  flutter_test:
    sdk: flutter

# The "flutter_lints" package below contains a set of recommended lints to
```

Add Library

Step7. Next, open the **main.dart** file. Then replace the contents with the script below. And don't forget to replace each **http://10.0.2.2/flutter_upload_image/** with the appropriate endpoints for each of your IP Address or use 10.0.2.2 IP Address.

```
import 'dart:convert';
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:image_picker/image_picker.dart';
import 'package:http/http.dart' as http;

void main() => runApp(MaterialApp(
  home: Home_Send_Get_Image_Roq(),
  debugShowCheckedModeBanner: false,
  theme: ThemeData(useMaterial3: true),
));

class Home_Send_Get_Image_Roq extends StatefulWidget {
  @override
  _Home_Send_Get_Image_RoqState createState() => _Home_Send_Get_Image_RoqState();
}

class _Home_Send_Get_Image_RoqState extends State<Home_Send_Get_Image_Roq> {
  XFile? image_roq;

  List _images_roq = [];

  final ImagePicker picker_roq = ImagePicker();

  //we can upload image from camera or from gallery based on parameter
  Future sendImage_roqFun(ImageSource media) async {
    var img = await picker_roq.pickImage(source: media);

    var uri = "http://10.0.2.2/flutter_upload_image/create.php";
```

```

var request = http.MultipartRequest('POST', Uri.parse(uri));

if (img != null) {
  var pic = await http.MultipartFile.fromPath("image", img.path);

  request.files.add(pic);

  await request.send().then((result) {
    http.Response.fromStream(result).then((response) {
      var message = jsonDecode(response.body);

      // show snackbar if input data successfully
      final snackBar = SnackBar(content: Text(message['message']));
      ScaffoldMessenger.of(context).showSnackBar(snackBar);

      //get new list images
      getImageServer_roqFun();
    });
  }).catchError((e) {
    print("CatchError of Create Opreation :$e");
  });
}

Future getImageServer_roqFun() async {
  try {
    final response = await http
      .get(Uri.parse('http://10.0.2.2/flutter_upload_image/list.php'));

    if (response.statusCode == 200) {
      final data = jsonDecode(response.body);

      setState(() {
        _images_roq = data;
      });
    }
  } catch (e) {
    print("Catch Error of get Image from Server: $e");
  }
}

@override
void initState() {
  // ignore: todo
  // TODO: implement initState
  super.initState();
  getImageServer_roqFun();
}

//show popup dialog
void myAlert_roqFun() {
  showDialog(
    context: context,
    builder: (BuildContext context) {
      return AlertDialog(
        shape:

```

```

        RoundedRectangleBorder(borderRadius: BorderRadius.circular(8)),
        title: Text('Please choose media to select'),
        content: Container(
          height: MediaQuery.of(context).size.height / 6,
          child: Column(
            children: [
              ElevatedButton(
                //if user click this button, user can upload image from gallery
                onPressed: () {
                  Navigator.pop(context);
                  sendImage_roqFun(ImageSource.gallery);
                },
                child: Row(
                  children: [
                    Icon(Icons.image),
                    Text('From Gallery'),
                  ],
                ),
              ),
              ElevatedButton(
                //if user click this button. user can upload image from camera
                onPressed: () {
                  Navigator.pop(context);
                  sendImage_roqFun(ImageSource.camera);
                },
                child: Row(
                  children: [
                    Icon(Icons.camera),
                    Text('From Camera'),
                  ],
                ),
              ),
            ],
          ),
        ),
      );
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Upload Image'),
        actions: [
          IconButton(
            onPressed: () => myAlert_roqFun(),
            icon: Icon(Icons.upload),
          ),
        ],
      ),
      body: _images_roq.length != 0
        ? GridView.builder(
            gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(
              crossAxisCount: 2),
            itemCount: _images_roq.length,
            itemBuilder: (_, index) {

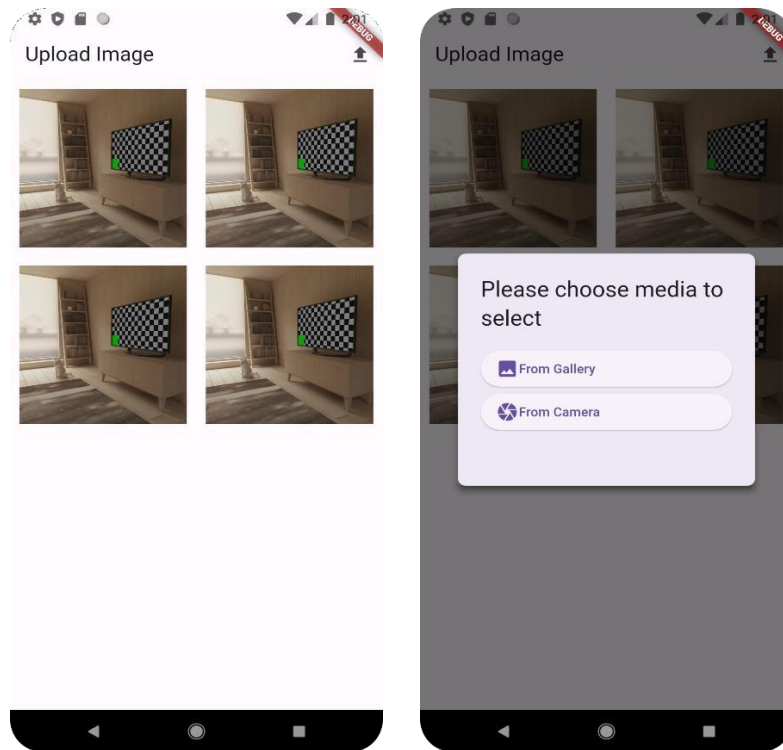
```

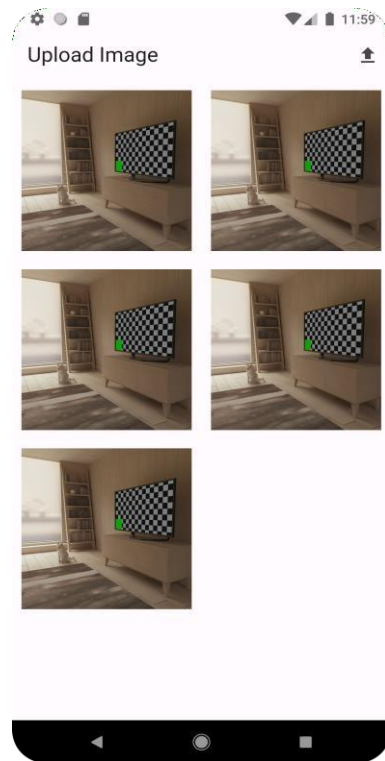
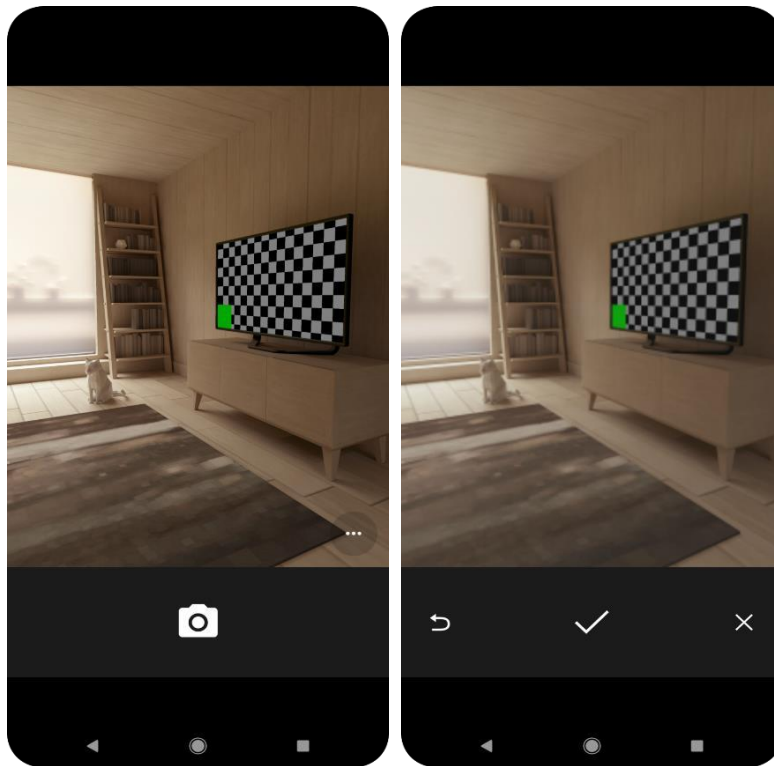
```

return Padding(
  padding: EdgeInsets.all(10),
  child: Image(
    image: NetworkImage(
      'http://10.0.2.2/flutter_upload_image/images/' +
        _images_roq[index]['image']),
    fit: BoxFit.cover,
  ),
);
})
: Center(
  child: Text("No Image"),
);
}
}

```

Step8. Ok, we have gone through all the steps, it's time to do a trial. Please run the emulator and run the project we just finished creating. If successful, it will look like the image below.





Final Result

